

Input and Output Specification

1. Main Inputs

A. Raw Datasets (Data Source Layer)

The system ingests transaction records from raw CSV datasets located in `Datasets/`:

Dataset	Path	Records	Columns / Features	Target Label
IEEE-CIS Fraud Detection <i>(Primary)</i>	<code>Datasets/IEEE CIS-.../IEEE CIS/train_transaction.csv</code>	590,540	434 columns	<code>isFraud</code> (0 or 1)
Credit Card Fraud <i>(Secondary)</i>	<code>Datasets/Credit card Fraud detection/credit_card_fraud_10k.csv</code>	10,000	10 columns	<code>is_fraud</code> (0 or 1)
PaySim Mobile Money	<code>Datasets/Paysim/paysim dataset.csv</code>	6,362,620	11 columns	<code>isFraud</code> (0 or 1)

B. Ingestion Stream Inputs (Kafka Producer)

- **Message Key:** `card1` or `card_id` (used for deterministic card-level partition routing).
- **Producer Configuration:**
 - Broker: `localhost:9092`
 - Replay rate: Configurable (`--rate` parameter, e.g., 1,000–2,000 tx/sec).
 - Batch size / count: Configurable (`--count` parameter).

2. Main Outputs

A. Kafka Ingestion Stream (Transport Output)

- **Topics:** `ieee_cis_transactions`, `transactions_p3`, `transactions_p6`, `transactions_p12`.
- **Serialization:** UTF-8 JSON.
- **Payload Modes:**
 - **Standard / Compact JSON:**

```
{
  "TransactionID": 2987000,
  "isFraud": 0,
  "TransactionDT": 86400,
  "TransactionAmt": 68.50,
  "ProductCD": "W",
  "card1": "13926",
  "card2": "150.0",
```

```
"card3": "150.0",  
"card4": "discover",  
"card5": "142.0",  
"card6": "credit",  
"addr1": "315.0",  
"addr2": "87.0",  
"P_emaildomain": "gmail.com",  
"R_emaildomain": "",  
"timestamp": 1791130000.123  
}
```

- **Full Payload Mode:** All 434 raw features (C1..C14, D1..D15, M1..M9, V1..V339).

B. Downstream Processing Outputs

1. Apache Spark (Micro-batch ML Inference & Analytics):

- **Feature Dataframe:** Real-time parsed columns with card-partitioned keys.
- **Model Inference Scores:** Fraud probability / risk predictions from LightGBM (E1).
- **Sink Targets:** Streaming Console output / Parquet storage sinks / Alert queues.

2. Apache Flink (Low-Latency Stream & CEP):

- **Velocity Alerts:** Flags for rapid transactions (e.g., > 3 transactions within 60 seconds per card).
- **Stateful Anomaly Indicators:** Sliding-window aggregations and rule-based fraud signals.